A: TGS + explicit PD

B: TGS + PhysX joint drive

C: PGS + PhysX joint drive

rotor inertia $(10^{-3}$ kg m$^2)$

theory: slope 1.00

fit: slope 2.41, R^2 = 0.9875

theory: slope 1.00

fit: slope -0.13, R^2 = 0.9613

theory: slope 1.00

fit: slope -0.20, R^2 = 0.8835

$h = 1/f$ (ms)

ΔK_D $(10^{-2}$ N m s/rad)

theory: slope 4.00

fit: slope 3.62, R^2 = 0.9606

theory: slope 4.00

fit: slope -1.85, R^2 = 0.9623

theory: slope 4.00

fit: slope -4.41, R^2 = 0.9938

ΔK_P (N m/rad)

theory: slope 0.00

fit: quadratic, R^2 = 0.0596

theory: slope 0.00

fit: quadratic, R^2 = 0.9999

theory: slope 0.00

fit: quadratic, R^2 = 0.9728

post-fit RMSE (rad)

theory: RMSE $\propto h^2$

h^2 (ms^2)

Isaac Sim 5.1 / PhysX. Parameter rows: first-order theory (solid), fit (dashed). Last row: post-fit RMSE against h^2.